MILITARY SPECIFICATION SHEET

AMPLIFIER, RESOLVER

The complete requirements for procuring the resolver amplifier described herein shall consist of this document and the issue in effect MIL-A-81258.

Scope - This specification covers the detail requirements for a transistorized amplifier, used in resolver systems using trimmed computing resolvers and low reactance a-c summing network.

Class - Operating temperature range -55° C to 85° C
Nonoperating temperature range -62° C to 125° C
Altitude 0 to 100,000 ft.

Physical Characteristics

Dimensions	See Figure 1
Weight	10.5 oz (max.)

NOTES: (a) The open loop tests shall be conducted as in Figure 2. When the load is not specified, load "A" Table I shall be used. The closed loop tests shall be conducted as in Figure 3.

(b) The DC power supply to be used shall have the following characteristics:

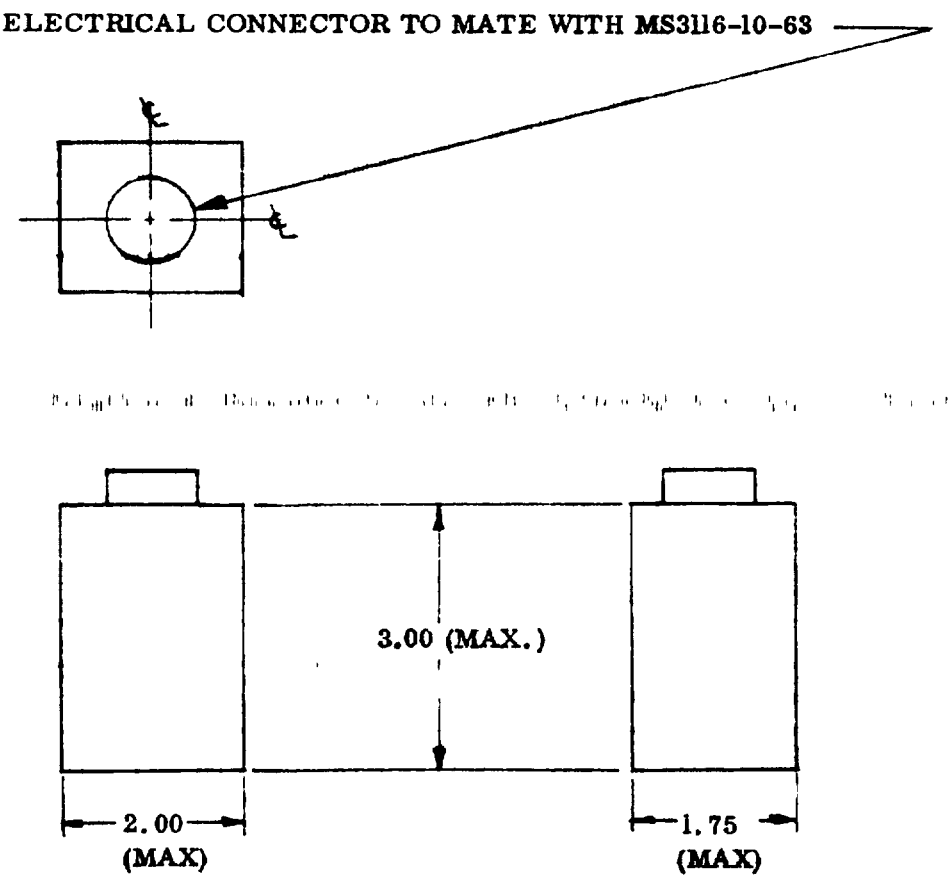
Output voltage	-53.5v±0.1v, -55v±0.3v, -56.5v±0.1v
Output current	700 ma (minimum)
Output impedance	1 + j0 ohms (maximum)
Ripple voltage	3mv peak to peak (maximum)

FSC 5990

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PIN

A	RESOLVER EXCITATION
B	INPUT
C	SIGNAL GROUND
D	POWER GROUND
E	DC POWER SUPPLY
F	SPARE

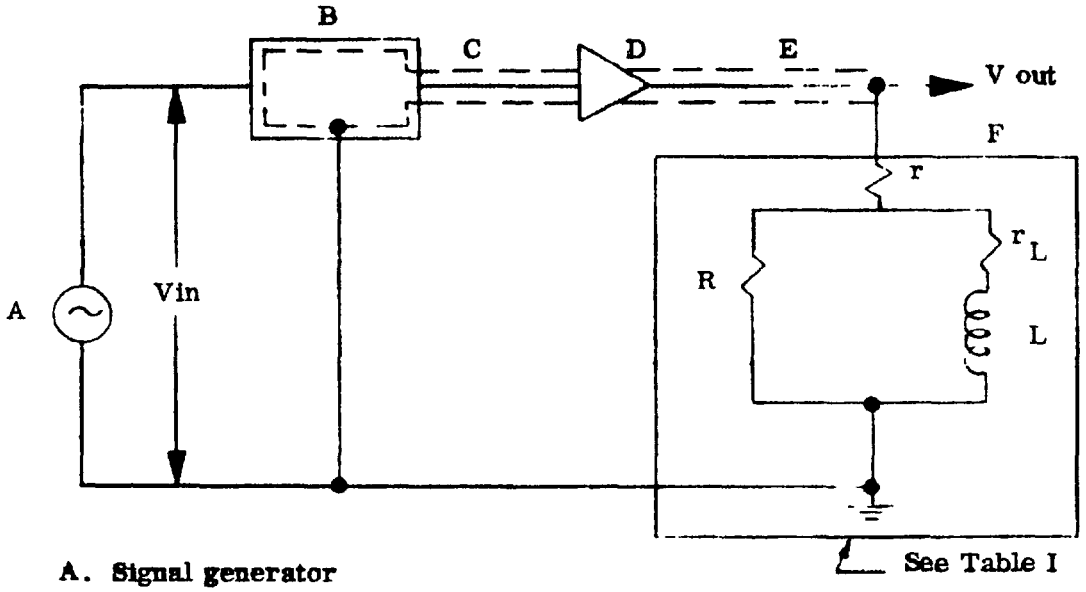


NOTE:

DIMENSIONS ARE IN INCHES.

FIGURE 1 RESOLVER AMPLIFIER, OVERALL DIMENSIONS

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A. Signal generator
B. Summing resistor network; (see Appendix) unused input terminals to be grounded.
C. Input 95 ohms coax cable; total capacitance 30pf±2pf
D. Resolver amplifier to be tested
E. Output 95 ohms coax cable, 10 feet maximum length
F. Dummy load.

Figure 2. Open Loop Test Circuit

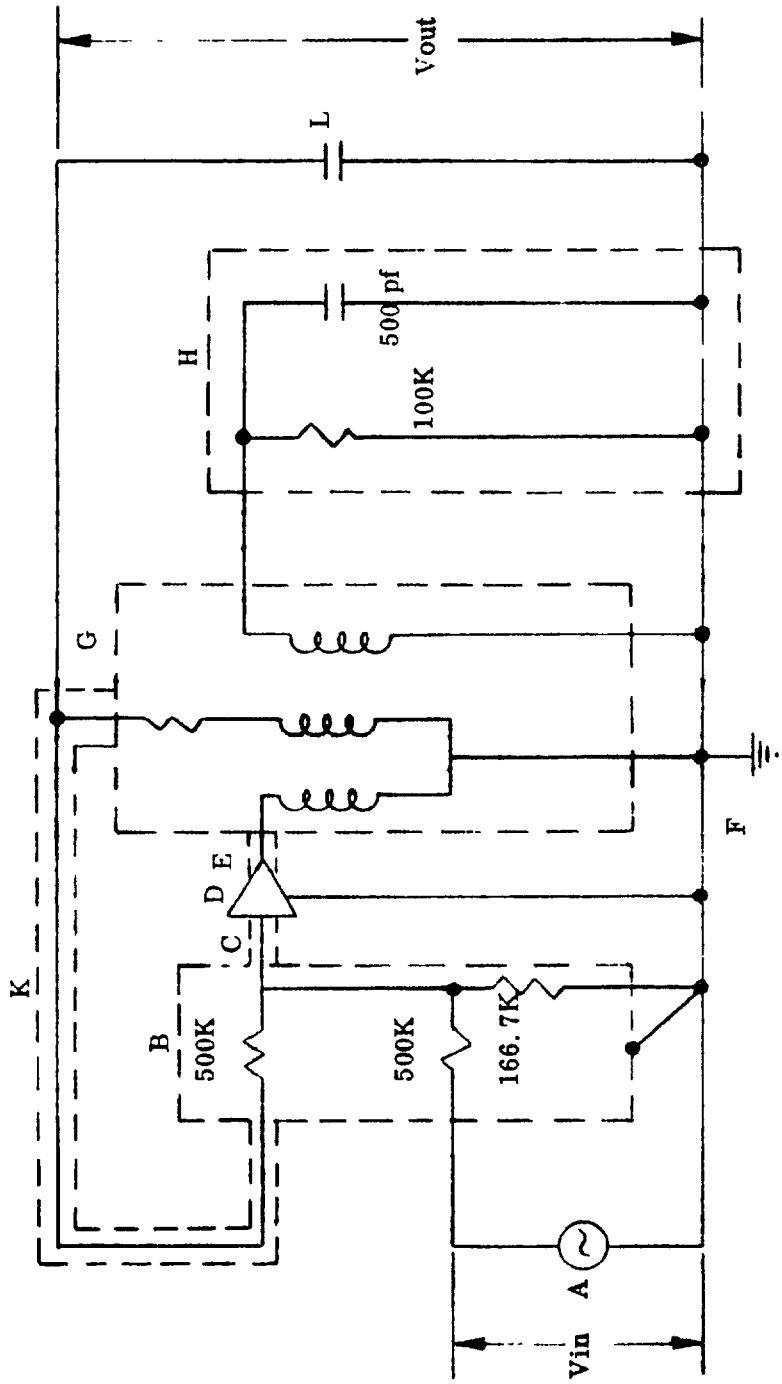
TABLE I

DUMMY LOAD DATA

LOAD	$r+r_L$ ±2% (ohms)	*r_L (ohms) (typical)	L±2% (Henries)	R±2% (ohms)
A	395	45	1.0	33K
B_1	310	40	0.9	27K
B_2	440	40	0.9	27K
B_3	555	40	0.9	27K
C_1	245	50	1.1	39K
C_2	380	50	1.1	39K
C_3	440	50	1.1	39K

*r_L is the resistance of the inductor L

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A. Signal generator
B. Summing resistor network (see appendix)
C. Input 95 ohms coax cable; total capacitance 30pf±2pf
D. Resolver amplifier to be tested
E. Output 95 ohms coax cable, 10 feet maximum length
F. Resistance of common ground lead between amplifier and resolver 0.25 ohm (maximum)
G. Trimmed resolver (see MIL-R-23417/)
H. Specified rotor loading
K. 95 ohms coax cable
L. Additional capacitance to the coax cable (K) to obtain the total capacitance load of 500pf

Figure 3. Closed Loop Test Circuit

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Test Conditions and Performance Characteristics

PARAGRAPH IN MIL-A-81258		Test Conditions or Performance Characteristic	Value
Par. in Section 3	Par. in Section 4		
3.4.8.1		Open loop - input voltage range	0 to 60 mv rms
		Open loop - nominal output voltage	20v rms
		Closed loop - input voltage range	0 to 26v rms
		Closed loop - nominal input voltage	20v rms
3.4.8.2	4.3.4.b	Signal generator output signal	26v rms 400 cps
	4.3.4.c	Signal generator output voltage	52v rms
	4.3.4.d	Repeat the test at	-55° C and 85° C
3.4.8.3	4.3.5.d	Maximum DC power supply current	37 ma
	4.3.5.b	Signal generator output signal	ov rms
	4.3.5c	Signal generator output voltage	52v rms
3.4.8.4		DC power input	1.75 ± .40 watts
	4.3.6.b	DC supply voltage	-55±1 volts
	4.3.6.c	Input signal	20v rms±5%, 400±15cps
	4.3.6.d	DC supply current	31.5±3 ma
3.4.8.5	4.3.7.c	Phase shift (at resonance)	180° ±10°
	4.3.7.b	Amplifier Output signal	20v rms, 400±15cps
3.4.8.6	4.3.8.d	Open loop input impedance	390K±10%
	4.3.8.b	Amplifier output signal	20v rms 400±15cps

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Test Conditions and Performance Characteristics (Cont'd)

PARAGRAPH IN MIL-A-81258		Test Conditions or Performance Characteristics	Value
Par. in Section 3	Par. in Section 4		
3.4.8.7	4.3.9	Not applicable	
3.4.8.8	4.3.10	Temperature and altitude Class 2 with the following exceptions:	
		Step 3	100,000 ft.
		Step 5	125° C
		Step 6	85° C
		Lowest input voltage	0
		Corresponding output voltage (max.)	1mv rms
		Highest input voltage	26 v rms
		Corresponding output voltage	26 v rms
		Step 14 - Input voltage	20 v rms
		Corresponding output voltage	20 v rms
	4.3.10.b	The parts shall be 4.3.13.e(parts 1 to 5 inclusive) and 4.3.13(parts (a) to (d) inclusive)	
3.4.8.9	4.3.11.c	**Internal noise (max.)**	1mv rms
3.4.8.10	4.3.12.c	**Warm-up time (max.)**	1.5 second
	4.3.12.b	Amplifier input signal	30mv rms, 400±15cps
3.8.11		Voltage gain and linearity	
	4.3.13.b	Amplifier input voltage	20 v rms
		Gain	.9994±.0002± j.0002
	4.3.13.c	Amplifier input voltage	30mv rms
			gain / frequency
		Lower -3db frequency	.707 / 5 cps (max.)
		Lower Resonance frequency	3 (max.) / 8 ± 1 cps
		Upper resonance frequency	2 (max.) / 13.5 ± 1.5 Kcps
		Upper -3db frequency	.707 / 35 Kcps (max.)

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Test Conditions and Performance Characteristics (Cont'd)

PARAGRAPH IN MIL-A-81258		Test Conditions or Performance Characteristics	Value
Par. in Section 3	Par. in Section 4		
	4.3.13.e	Signal Generator output voltage	30 mv rms
		frequency (cps) / Min. input voltage saturating the amplifier (V rms)	
	4.3.13.g	400±15 / 30.0	
	4.3.13.h	800±30 / 12.1	
		1200±45 / 6.8	
		2400±60 / 2.9	
		3600±60 / 1.8	
	4.3.13.j	Voltage gain and linearity (open loop)	

1. Set the following

DC Power Supply	Output V rms	freq. (cps) (resonance)	Load	temp. °C
-56±.3V	20±.5	400±25	A	-55

2. Measure the input voltage and compute the gain.
3. The gain in part (2) shall be 2800±500; The bandwidth shall be 105±10 cps.
4. Repeat parts (1) and (2) for temperatures at 20° C and 85°C
5. The gains in part (4) shall be 3200±500. The resonance frequency shall be 400±15cps and 400±25 cps respectively. The bandwidth shall be 105±10 cps.
6. Change the load to B_1, B_2, and B_3, the temperatures to -55°C, 20°C and 85°C respectively and the frequency to 380 ± 2 cps. At each setting perform part (2).
7. Each gain in part (6) shall exceed 1500.
8. Change the load to C_1, C_2, and C_3, the temperatures to -55°C, 20°C and 85°C respectively, the DC power supply to -53.5±0.1 V and the frequency to 400±25cps(resonance). At each setting perform parts (9a) and (9b).

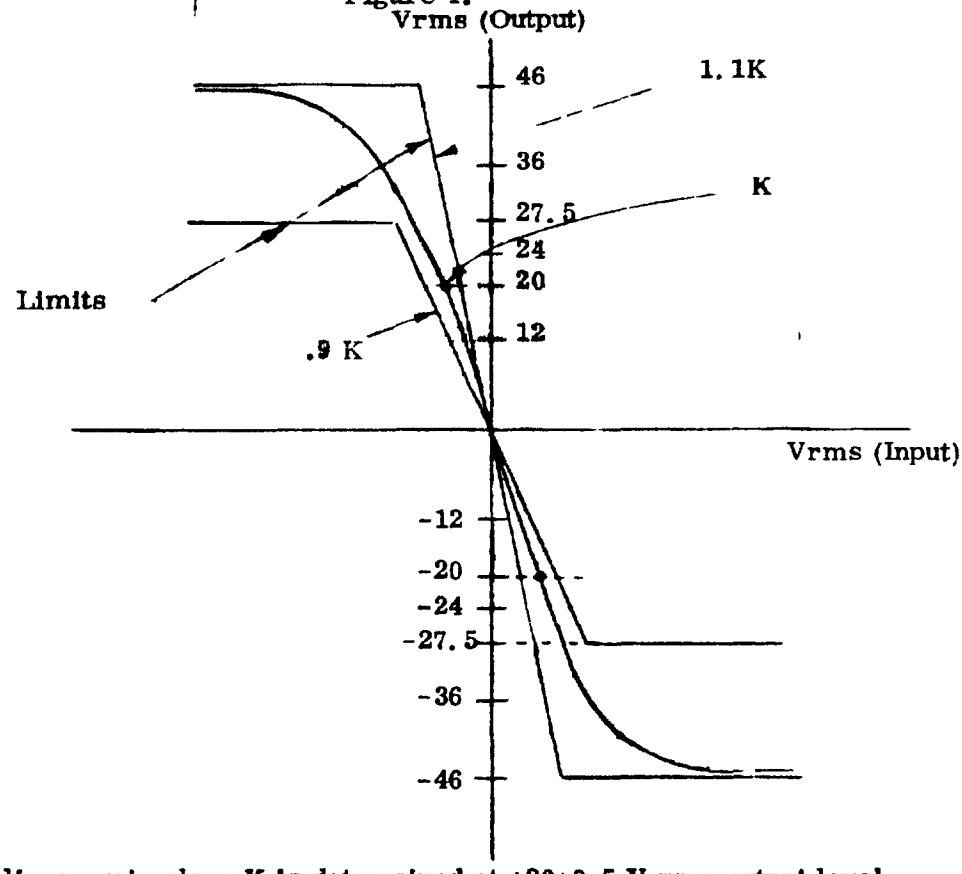
9a. Change the input signal slowly from 0 V rms to 60 MV rms. There shall be no instability, oscillations or any other peculiar behavior of the amplifier.

9b. Add 75 ohms (min.) in series with 30 mhy (min.) to the output of the power supply and repeat part (8). (This addition shall be considered as a part of the output impedance of the power supply).

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Test Conditions and Performance Characteristics (Cont'd)

PARAGRAPH IN MIL-A-81258		Test Conditions or Performance Characteristic
Par. in Section 3	Par. in Section 4	
	4.3.13.j	10. Vary the DC power supply to -55± .3V and -56.5±.1V. At each level repeat parts (8). (9a) and (9b). 11. Vary the DC power supply to -53.5±.1V and the output impedance as in part 9b, the temperature to 20° C, Load B_2 and obtain data to plot the output voltage versus input voltage (in the range from 0 to 60 mv rms). The frequency shall be 380±2cps. 12.* The plot of part (11) shall be within the limits of Figure 4.



* The linear gain slope K is determined at ±20±0.5 V rms output level

Figure 4. Open Loop Gain Plot

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Test Conditions and Performance Characteristics (Cont'd)

PARAGRAPH IN MIL-A-81258		Test Conditions or Performance Characteristic	Value
Par. in Section 3	Par. in Section 4		
3.4.8.12	4.3.14	The performance test shall be in accordance with paragraph 4.3.13.	
3.4.8.13	4.3.15	Same as in paragraph 3.4.8.12.	
3.4.8.14	4.3.16	The performance test shall be in accordance with paragraph 4.3.13.e (parts 1 to 5 inclusive) and 4.3.13. (parts (a) to (d) inclusive)	
3.4.8.15	4.3.17	Same as in paragraph 3.4.8.14	
3.4.8.16	4.3.18	Same as in paragraph 3.4.8.14	
3.4.8.17	4.3.19	Same as in paragraph 3.4.8.14	
3.4.8.18	4.3.20	Same as in paragraph 3.4.8.12	
3.4.8.20	4.3.22	Input signal	20 V rms 400±15cps
		Output voltage	20 V rms
		Altitude	100,000 ft.
		Accept-reject criteria Figure 11 (MIL-R-23094)	

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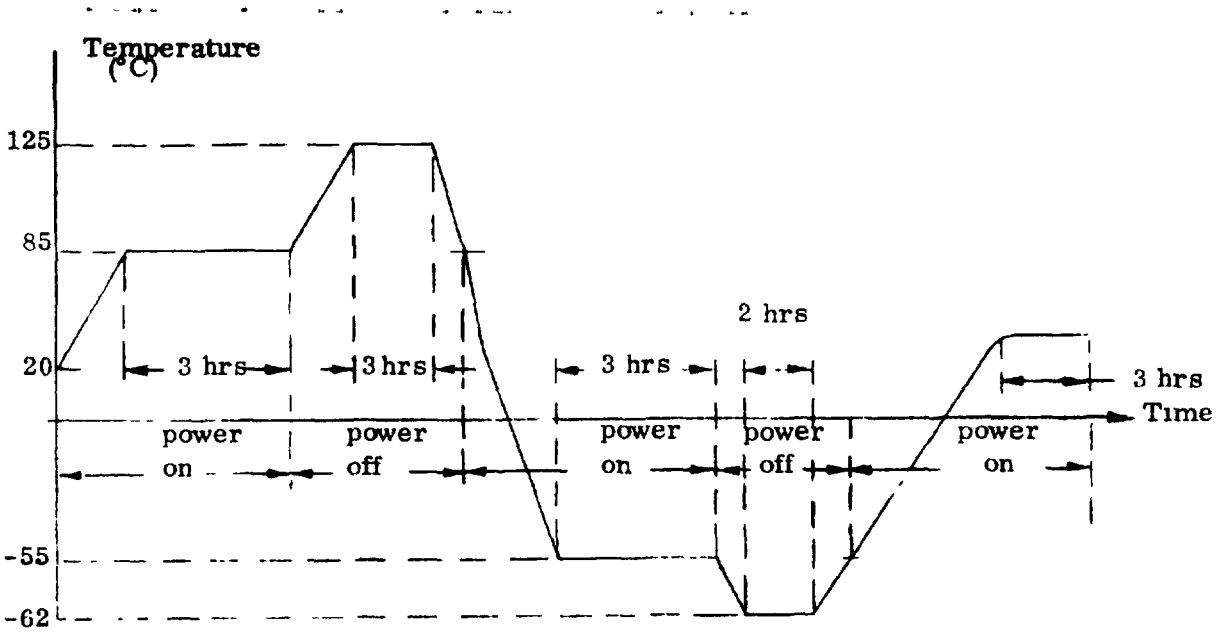


Figure 5. Reliability Test Cycle

NOTE: Heat portion and cool portion of the cycle could be done in alternate days.

Identification: This amplifier identified as AM-3494/U during development, shall be hereafter identified by part number M81258/1-001.

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APPENDIX

Summing Network - The characteristics of the summing network described herein are based on the consideration that the network is a separate entity capable of being defined and tested. The physical form or location of the network is not restricted.

a) Resistors

b) The DC resistance of each resistor shall not deviate from the specified values by more than .02%.

c) The temperature coefficient of each resistor shall not exceed 10 ppm/°C.

d) Each resistor shall be considered as having an equivalent circuit of a resistor and capacitor in parallel. The time constant of the resistor, when connected into, and operating as part of, the network shall not exceed 0.2 μ sec.

e) Stray Capacitance - The capacitance between each terminal connection and shield shall not exceed 25 pf.

f) Transfer Ratio - The transfer ratio, $TR_{12}=\frac{E_1}{E_2}$, shall be according to Figure 6; for $E_3=0$.

CONDITION #1
Temp = Room temp. ambient
Altitude = Sea Level
Input Signal = 20 V $\pm$ 1 V RMS, 400 cps $\pm$ 5%
TR = $-1 \pm 0.0002 \pm j0.0001$

CONDITION #2
Temp range = -55° C to +85°C
Altitude = Sea level to 100,000 ft.
*Input Signal = 20 V $\pm$ 1 V RMS, 400 cps $\pm$ 5%
TR = $-1 \pm 0.0005 \pm j0.0003$

g) If it has been demonstrated that the circuit described herein meets the requirements of CONDITION #2 then CONDITION #1 may be used as a process check.

h) Reliability - Reliability of the circuit shall be determined by successfully withstanding the duty cycle described in 3.4.8.20 for an MTBF of 10,000 hours. The input signal shall be 20 V $\pm$ 1 V (RMS) at 400 cps $\pm$ 5%.

* Transient not to exceed 52 V (RMS)

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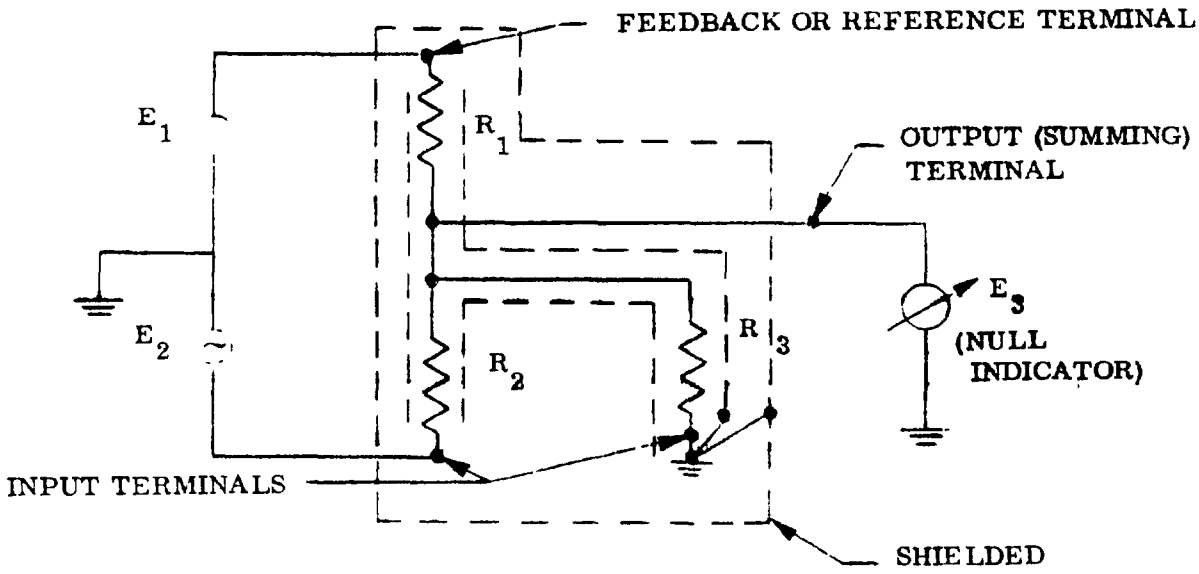


Figure 6. Transfer Ratio Test Setup

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SPECIFICATION ANALYSIS SHEET

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INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION
MIL-A-81258/1(WP) AMPLIFIER, RESOLVER

ORGANIZATION *(of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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